Chowdhury Md Rajib Sarwar

Senior iOS Engineer

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Professional Summary

Senior iOS Engineer with 10+ years of experience architecting resilient, high-performance mobile systems across enterprise and consumer domains. Specializing in offline-first synchronization, concurrent data pipelines, legacy modernization, and low-level device integrations. Currently leading mobile architecture for a highly scalable white-label ecosystem powering 200+ client applications.

Technical Skills

Languages & Core iOS: Swift, Objective-C, Kotlin, SwiftUI, UIKit, Cocoa Touch, C++ (Interoperability)
Architecture & Systems: System Design, Server-Driven Configuration, MVVM, Clean Architecture, Dependency Injection, Protocol-Oriented Design, Offline-First Architecture
Concurrency & Performance: Swift Concurrency (Actors, Task Groups, async/await), GCD, Memory Management, Xcode Instruments (Time Profiler, Leaks), Render Loop & Main-Thread Optimization
Networking & Data: REST APIs, WebSockets / Raw Sockets, Core Data, Distributed Caching, Background Synchronization, WKWebView / JavaScript Bridging
Media & Hardware Integrations: AVFoundation, A/V Rendering Pipelines, Core Location (High-Frequency GPS), Bluetooth / Hardware SDKs, Push Notifications
Testing, Quality & DevOps: XCTest, XCUITest, Test-Driven Development (TDD), CI/CD Pipelines, Diagnostic Logging, Observability, Advanced Crash Analysis

Professional Experience

Senior Software Engineer, iOS — Limosys LLC **Jan 2023–Present**
Englewood Cliffs, NJ

- Architected a dynamic, server-driven configuration pipeline to replace a fragmented, hard-coded white-label ecosystem, reducing new client application deployment steps by 75% (from 20 to 5) across 200+ enterprise apps.
- Spearheaded the modernization of legacy Objective-C MVC monoliths into a modular Swift MVVM architecture, integrating secure WKWebView JavaScript bridges and migrating decentralized client assets to a unified Amazon S3 infrastructure.
- Engineered an offline-first data synchronization layer for the driver application, utilizing local database persistence and background processing to ensure zero data loss for critical GPS tracking and signature captures during network outages.
- Rescued a stalled third-party hardware integration by collaborating directly with Worldpay engineers to debug and stabilize a Bluetooth-connected triPOS SDK, implementing comprehensive diagnostic logging to successfully deploy in-car payments.
- Currently architecting a high-frequency real-time GPS map-matching pipeline utilizing GraphHopper and Hazelcast in-memory computing to process 10-second location pings and snap raw coordinate data to road networks for precise fleet routing.
- Designed and implemented a modernized Android counterpart utilizing Kotlin, Jetpack Compose, and MVVM, establishing cross-platform architectural parity and stability across the driver ecosystem.

Mobile Programmer / Team Lead — New Jersey Institute of Technology **Sep 2021–Dec 2022**
Newark, NJ

- Directed a cross-functional engineering team (iOS, Android, UI/UX) to architect and launch "The DANA Project" from zero-to-one, earning the 2023 NJLA Technology Innovation Award for outstanding digital accessibility.
- Engineered a secure data-ingestion layer bridging siloed third-party APIs and institutional backend systems, ensuring strict data privacy and continuous state synchronization for the native mobile clients.
- Mentored developers on advanced UIKit implementations, guiding technical decisions around iOS memory management, view lifecycles, and state preservation to guarantee a highly responsive, crash-free App Store release.
- Acted as the primary technical liaison between university stakeholders and the engineering team, successfully translating complex web-archival constraints into a scalable, location-aware mobile architecture.

Engineering Manager, Mobile — Affle (Acquired Shoffr) **Jan 2017–Sep 2021**
Singapore / Bengaluru, India

- Led end-to-end mobile product architecture for the Shoffr omnichannel (O2O) platform, partnering directly with the C-suite and external retail clients to deliver an iOS application that directly contributed to a successful Techstars seed funding round and eventual acquisition by Affle.
- Engineered robust offline-first synchronization workflows and real-time customer-to-retailer communication protocols, enabling seamless in-store footfall attribution and inventory discovery through localized Arrival Code scanning.
- Architected and deployed the proprietary Vizury mobile SDK across Affle's consumer intelligence network, implementing highly efficient background task management and offline data caching to track user behavior and ad impressions at scale.
- Designed GDPR-compliant data collection and state management pipelines within the SDK, optimizing on-device resource constraints (battery and CPU) while enabling targeted omnichannel re-engagement via push notifications and messaging.

Software Engineer, iOS — AnyConnect **Oct 2014–Jul 2016**
Dhaka, Bangladesh / Singapore

- Stabilized the Philips uGrow application architecture by auditing and overhauling A/V rendering pipelines and memory management, resolving 1,000+ critical bugs to ensure continuous, crash-free video streaming.

- Architected resilient networking layers utilizing persistent XMPP protocols and raw socket connections, significantly reducing stream latency and connection drops between the mobile client and IoT camera hardware.
- Engineered efficient background data synchronization via push notifications to accurately maintain application state, while unblocking the main thread to restore 60fps UI performance.
- Recognized for critical technical impact on the Philips project with a direct promotion and relocation from the Dhaka engineering center (Eyeball Networks) to the Singapore headquarters (AnyConnect).

Selected Projects

LSN Driver App | *Swift, UIKit, GPS, Payments, REST APIs* | App Store

- Engineered a mission-critical enterprise driver application supporting continuous, high-frequency GPS tracking, secure payment gateways, and bidirectional dispatch workflows.

Qibrah | *Swift, SwiftUI, Speech Recognition, Live Activities* | App Store

- Architected and independently launched a comprehensive iOS utility application, utilizing a strict SwiftUI and MVVM pattern to deliver highly responsive, state-driven interfaces.
- Integrated AVFoundation for resilient audio playback, pairing it with modern iOS Live Activities and push notifications to maintain continuous background state and user engagement.
- Engineered an on-device Speech-to-Text pipeline utilizing Apple's Speech framework, implementing a custom real-time algorithm that synchronizes UI text highlighting with live audio progression.
- Designed a robust local persistence layer to manage complex application states, ensuring offline-capable session continuity for user reading progress and bookmarks.

Honors & Awards

NJLA CUS/ACRL-NJ Technology Innovation Award — June 2023
Awarded by the New Jersey Library Association College and University Section (C&US) and the ACRL New Jersey Chapter in recognition of technology innovation and impact; associated with **The DANA Project**.

Education

New Jersey Institute of Technology — **Sep 2021–Dec 2022**
Master of Science in Computer Science — *New Jersey, USA*

Ahsanullah University of Science and Technology — **Sep 2010–Aug 2014**
Bachelor of Science in Computer Science — *Dhaka, Bangladesh*